



M003 Week 5 — English Worksheet (Beginner)

Topic: Layer Up — Trees · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

You know the floor — **(x, z)**. This week you go **up**. One **fill** makes a solid box between **two corners**. Raise the **middle number (y)** and the box gets **taller**. Stack a wide box, then smaller boxes on top, and you get a **tree**.

Your mission: build this tree — a fat brown **trunk**, then a green **cap** of boxes that get **smaller** going up.

Build this



The idea

- A **3 × 3** brown **trunk**, stacked up.
- A green **cap**: bottom box **widest**, each box above **smaller and higher**.
- Each box is one **fill**.

Every coordinate is **(x, y, z)**

1st = X

across →

2nd = Y

up ↑ (height)

3rd = Z

forward ↗

Big idea: the **2nd number (y)** is height. Make it bigger to go **up**.

1 · Predict 🧠

Read the steps. Look at each **column** — **1st = x, 2nd = y, 3rd = z**. Circle or write your answer.

```
fill BROWN from (3, 0, 3) to (5, 6, 5)
```

The 2nd number (y) goes from 0 up to 6. Does this build the trunk going up? Circle one: yes · no

Why?

fill GREEN from (1, 7, 1) to (7, 8, 7)

The x numbers go 1 → 7 and the z numbers go 1 → 7. Is this a wide box? Circle one: yes · no

2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version. Circle what's different and write one short sentence about the bug.

Pair A — The second box should sit **on top**, so its **2nd number (y)** must be bigger.

```
# clean
fill GREEN from (1, 7, 1) to (7, 8, 7)
fill GREEN from (2, 9, 2) to (6, 11, 6)
```

```
# buggy
fill GREEN from (1, 7, 1) to (7, 8, 7)
fill GREEN from (2, 7, 2) to (6, 8, 6)
```

What is wrong? Do the two boxes stack up, or land on the same level?

Pair B — The trunk should go **up**, so the **2nd number (y)** must grow.

```
# clean
fill BROWN from (3, 0, 3) to (5, 6, 5)
```

```
# buggy
fill BROWN from (3, 0, 3) to (5, 0, 5)
```

What is wrong? Does the buggy one go up, or stay flat on the ground?

3 · Fill the Gap

The second cap box should sit **higher** than the first. The missing number is the **2nd one (y)**. Fill it in using the word bank.

```
fill GREEN from (1, 7, 1) to (7, 8, 7)
fill GREEN from (2, ____, 2) to (6, 11, 6)
```

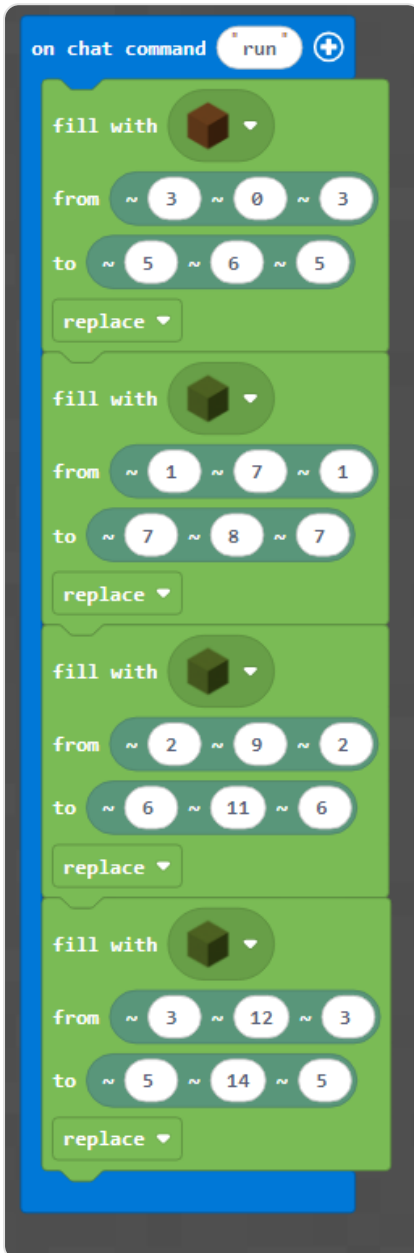
Word bank: 9 · 7 · 0

Write the missing number:

4 · Build the Tree 📷

Switch to your homework world and **build the tree from the top of this sheet**. Drag the blocks to match this exactly. Remember the colours: **1st = x** · **2nd = y** · **3rd = z**.

🧱 Blocks



1 2 3 4 The four boxes

Trunk (brown): from (3, 0, 3) to (5, 6, 5)

Cap 1 (green): from (1, 7, 1) to (7, 8, 7)

Cap 2 (green): from (2, 9, 2) to (6, 11, 6)

Cap 3 (green): from (3, 12, 3) to (5, 14, 5)

Send a **photo or video** of your tree, then explain what you did. Write 2 or 3 sentences.

My trunk is ... tall (the **y** number went up to ...).

The number that grew for each box was **y**, which means ...

My cap gets smaller because each box's **x** and **z** ...

My trunk is ... tall (the **y** number went up to ...).

The number that grew for each box was **y**, which means ...

My cap gets smaller because each box's **x** and **z** ...

My trunk is ... tall (the **y** number went up to ...).

The number that grew for each box was **y**, which means ...

My cap gets smaller because each box's **x** and **z** ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while you show your tree. Talk like you are teaching a friend. Try to use these words: **fill**, **box**, **x**, **y**, **z**, **trunk**, **cap**.

1. Show your tree from the bottom up to the top.
2. Point at one cap box and say its **y** number (how high it is).
3. Show two boxes and say which one is **wider**.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.